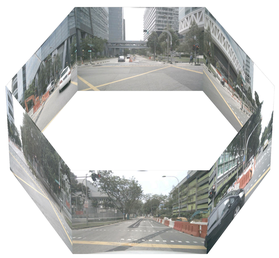
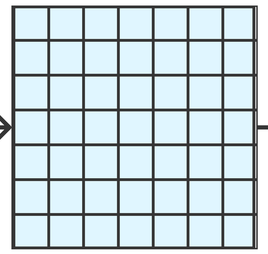
(a) Pretraining with Perception Tasks

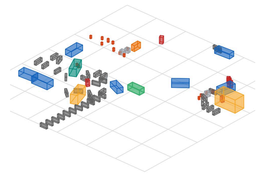
Multi-view Images



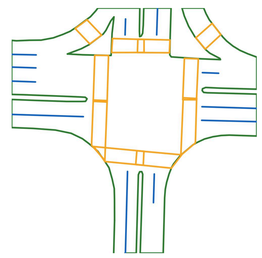
BEV Encoder

BEV Feature

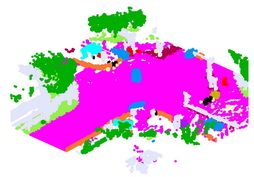




3D Object Detection



HD Map Prediction

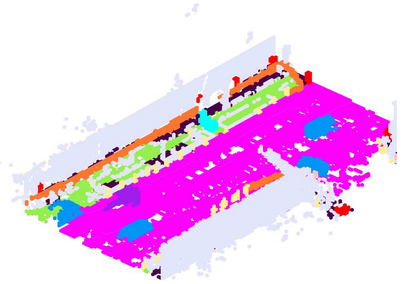


3D Occupancy Prediction

(b) Pretraining with Dense Reconstruction

Multi-view Images

Large Scale Encoder



3D Voxels

RGB Image

Depth Map

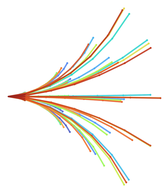
(c) Pretraining with Sparse Action Targets (Ours)

Safe

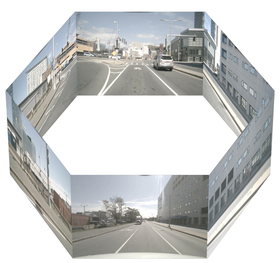
Risk

Unknown

Sparse Action Targets

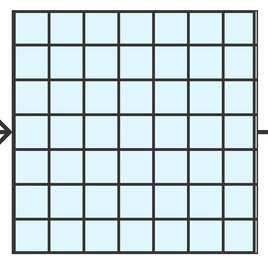


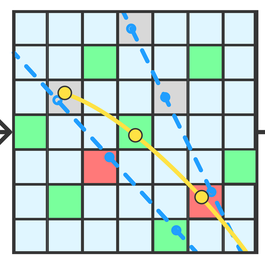
Multi-view Images



BEV Encoder

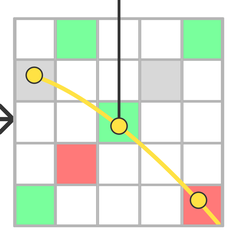
BEV Feature





Soft Labeling

Safe: 0.4 Risk: 0.4 Unknown: 0.2



Risk/Unknown Prediction